

Vaswani et al

Problem:

For time $t \in \mathbb{R}$, scalar representation in models is too weak. We need a multi-resolution representation that is complex enough

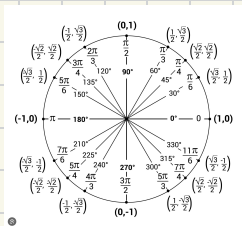
$$PE(t)_{2i} = \sin\left(\frac{t}{10,000^{2i/3}}\right)$$

$$PE(t)_{2i+1} = \cos\left(\frac{t}{10,000^{2i/3}}\right)$$

Modern way

$$emb(t) = \left\{ [\sin(\omega_k t), \cos(\omega_k t)] \right\}_{k=1}^{d/2-1}$$

each pair sin/cos is a unit circle



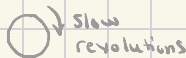
$$\omega_k = \exp\left(\frac{k}{d/2-1} \log 10,000\right)$$

$k: 0 \rightarrow d/2-1$

ω_k is a frequency

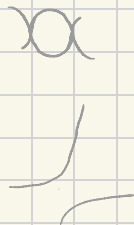
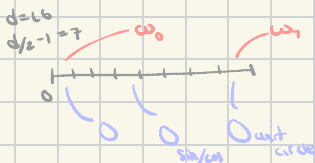
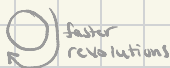
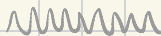
Low ω_k

larger wavelengths



High ω_k

smaller wavelengths



$$\text{Let } d=16 \rightarrow \omega_k = \exp\left(\frac{k}{7} \log 10,000\right)$$

$$k/7 \cdot \log 10,000 \approx [0, 1.3, 2.6, 3.9, 5.2, 6.5, 7.8, 9.2]$$

$$\exp(\cdot) \approx [1, 3.7, 14, 52, 193, 718, 2670, 10,000]$$

Now, we need both $\sin(\omega_k)$ and $\cos(\omega_k)$ to have complete information

Geometric Spacing & Measurement Scales

But what does ω_k do?

Essentially, we are creating a measurement system on the fly, tailored to the number of dimensions d , that wastes no information

$t = 1 \rightarrow 10$
1 2 3 4

Ex say $\omega_k = [1, 2, 4, 8, 16, 32, 64]$ $\omega_{k+1} = \omega_k r$

↑
unit of measurement

r is a scaling factor of the unit size between dimensions

ruler, ω_0

ruler, ω_1

⋮

ruler, ω_4

$\sin(\omega_1) \cos(\omega_1)$

$\sin(\omega_2) \cos(\omega_2)$

Once we have ω_k 's, our measurement system all we need is to pass in t 's to get embeddings

$$\text{emb}(t) = \{ [\sin(\omega_k t), \cos(\omega_k t)] \}_{k=0}^{d/2-1}$$

$$t \cdot \omega_k = t * (1, 3.7, 14, 52, 193, 718, 2670, 10000)$$

Why $e^{\log 10000}$? Why not just linear?

Let $d=8$



[1, 1429, 2857, 4286, 5714, 7143, 8571, 10000]

all the same scale

[1, 1429, 2857, 4286, 5714, 7143, 8571, 10-000]

t=1 * [1, 1429, 2857, 4286, 5714, 7143, 8571, 10-000]

t=2 * [1, 1429, 2857, 4286, 5714, 7143, 8571, 10-000]

t=10 * [1, 1429, 2857, 4286, 5714, 7143, 8571, 10-000]

(, , , , )

(, , , , )

...

(, , , , )

